DELTA EXCHANGE MULTI-AGENT TRADING SYSTEM

Complete End-to-End Architectural & Operational Documentation

Version: 2.5.0 Institutional **Architecture:** 10 Agents / 6 Layers **Exchange:** Delta Exchange India **R:R Target:** ≥ 2.5:1

1. Executive Overview & Design Philosophy

The **Delta Exchange Multi-Agent Trading System** is an institutional-grade algorithmic trading architecture designed specifically for crypto perpetual swaps and futures. Rather than relying on a single fallible model or rigid rule set, the platform divides the full quantitative trading lifecycle among **10 specialized autonomous AI agents** operating across **6 hierarchical layers**.

CORE OPERATIONAL TENETS

- **Asymmetric Risk/Reward (≥ 2.5:1)**: Every trade proposal must mathematically target at least 2.5 times the risked capital.
- **Strict Sniper Selectivity (Floor ≥ 85/100)**: 'Take fewer trades, but higher-conviction trades for maximum net profit.' Zero trades in neutral chop.
- **Mandatory Bracket Protection**: Every order placed on Delta Exchange is submitted with simultaneous, hard Stop Loss and Take Profit child orders.
- **Milestone Trailing Stop Schedule**: Eliminates the 'breakeven trap' by cutting downside risk by 60% at +0.8R (leaves 1.2R+ cushion), then locking +0.3R green profit at +1.1R.
- **Continuous Self-Auditing Memory**: Every closed trade is reflected on, graded (A+ to F), and used to update persistent pattern win/loss memory.

2. System Architecture & 6-Layer Hierarchy

The system enforces strict directional separation of concerns. Data and decisions flow through six sequential layers:

Layer	Component / Agents	Primary Core Function
Layer 1 Market Intelligence	• MarketDataValidator • Agent 1: RegimeAnalyst • Agent 2: ExposureCoach	Validates live feed sanity, maps 6-timeframe structure (15m to Monthly), detects SMC liquidity sweeps, and scores cross-asset market breadth health (0-100).

Layer 2 Alpha & Validation	• Agent 3: StrategyAgent • Agent 4: TimeMachineAgent	Generates IVL setups scoring ≥ 85/100; forward-simulates setups across 400 historical candles; verifies positive expected value (EV ≥ +0.35R).
Layer 3 Risk Governance	• Agent 5: RiskGatekeeper • RiskManager Engine	Enforces fixed dollar risk ($2-$4 USD), conviction multipliers (up to 2.0x for A+ setups), max 2 correlated crypto positions, and liquidation buffer (≥ 3x SL).
Layer 4 Execution & Management	• Agent 6: ExecutionAgent • Agent 7: TrailingStopManager • Agent 8: PortfolioMonitor	Submits bracket limit orders with idempotency; manages multi-stage trailing SL (+0.8R risk cut, +1.1R green lock, +1.5R, +2.5R TP); reconciles journal fills.
Layer 5 Learning & Optimization	• Agent 9: AdaptiveLearningEngine • Agent 10: PerformanceCoach	Conducts post-trade post-mortem reflection; grades trades (A+ to F); reinforces winning patterns (+10 pts) and blacklists failing ones; audits rule degradation.
Layer 6 Infrastructure & Protocol	• Orchestrator • DeltaExchangeClient • DeltaMCPServer • TelegramBot	Direct HMAC-SHA256 REST API communications; Model Context Protocol (MCP) server; 2-way Telegram confirmation approval buttons; emergency killswitch.

3. Detailed Tasks of Every Agent & Engine

Agent 1: Multi-Timeframe Regime Analyst

Source: multiagent_team/regime_analyst.py

- Synchronizes 6 distinct timeframes (15m, 1H, 4H, Daily, Weekly, Monthly) to eliminate single-timeframe bias.
- Calculates EMA ribbons (9, 21, 50, 200), RSI(14), ATR(14), MACD(12,26,9), and ADX(14) trend strength.
- Detects Smart Money Concepts (SMC): Buy-Side Liquidity (BSL) and Sell-Side Liquidity (SSL) pivot sweeps.
- Validates Market Structure Shifts (MSS) to ensure displacement confirms reversals before entering.
- Classifies candlestick patterns (Pinbars, Engulfing, Doji) and chart patterns (Double Tops/Bottoms, Trendline Slopes).

Agent 2: Exposure Coach & Cross-Asset Breadth

Source: multiagent_team/exposure_coach.py

- Computes universe Market Breadth: % of assets trading above their 21-period EMA.
- Monitors perpetual swap funding rates to detect extreme crowd crowding and squeeze risks.
- Synthesizes cross-asset metrics into a deterministic Market Health score (0-100).
- Determines Market Posture: AGGRESSIVE_EXPANSION (Health ≥ 75), MODERATE_SELECTIVE (50-74), DEFENSIVE_CAUTION (32-49), CASH_PRIORITY (< 32, zero new trades authorized).
- Engages RECOVERY_POSTURE after ≥ 2 consecutive stop-outs, slashing risk to 50% ($2.00 max risk) and clamping concurrency to 1.

Agent 3: Alpha & Strategy Generator

Source: multiagent_team/strategy_agent.py

- Evaluates setups against hard veto gates: major economic news, wide bid-ask spread (> 0.15%), or price drift (> 0.50%).
- Prioritizes the battle-tested Institutional Liquidity Sweep & Absorption (IVL) strategy.

- Enforces strict Sniper Quality Floor: minimum score ≥ 85/100 for primary assets (BTC, ETH, XAU) and ≥ 90/100 for secondary altcoins.
- Generates trade setup specifications with entry price, structural stop loss, and take profit targeting ≥ 2.5:1 R:R.

Agent 4: Time Machine Probability Engine

Source: multiagent_team/time_machine_agent.py

- Simulates proposed setups backwards across 400 historical candles under identical market conditions.
- Deducts realistic institutional friction: taker fees (0.10% round-trip), slippage (0.06%), and funding drag.
- Calculates the Wilson Score 95% Confidence Interval lower bound to prevent small-sample statistical illusions.
- Enforces an empirical expected value hurdle: requires EV ≥ +0.35R and historical win rate ≥ 70%.

Agent 5: Risk & Discipline Gatekeeper

Source: multiagent_team/risk_gatekeeper.py

- Applies a fixed dollar risk sizing model ($2.00 to $4.00 USD), scaling to 75%-100% available capital when safe.
- Executes Conviction Sizing: Tier 1 (A+ Prime Conviction) gets up to 2.0x contract boost on primary assets.
- Enforces Correlated Exposure Limits: maximum 2 open positions on the same side of correlated crypto assets.
- Verifies Liquidation Armor: confirms liquidation price is at least 3x farther than stop loss distance.
- Applies Drawdown Scaling: automatically scales position sizing down by 30% if drawdown exceeds 2%.

Agent 6: Execution & Orderbook Engine

Source: multiagent_team/execution_agent.py

- Translates approved trade decisions into signed bracket orders on Delta Exchange.
- Applies idempotency tokens (client_order_id) to prevent duplicate execution during network retries.
- Submits limit entry with slippage guards alongside native Stop Loss and Take Profit child orders.
- Executes post-fill verification: if an entry fills without a confirmed Stop Loss order, it immediately submits an emergency protective stop.

Agent 7: Trailing Stop Loss & Profit Sentinel

Source: src/trailing_stop_manager.py

- Continuously monitors live position mark prices against entry price and risk units.
- Milestone 1 (+0.8R to +0.9R): Slashes risk by 60% (moves SL to -0.40R), leaving 1.2R+ breathing space to avoid breakeven traps.
- Milestone 2 (+1.1R to +1.2R): Locks in guaranteed positive green profit (+0.30R), ensuring trade cannot exit at $0 breakeven.
- Milestone 3 (+1.5R to +1.6R): Advances stop to lock +0.85R profit.
- Milestone 4 (+2.5R to +2.6R): Executes full Take Profit target.
- Engages High-Volatility Buffer: widens trailing thresholds during expansion regimes to prevent stop hunting.

Agent 8: Portfolio Monitor & Memory Journaler

Source: multiagent_team/portfolio_monitor.py

- Polls active Delta Exchange positions, liquidation margins, and unrealized PnL.
- Reconciles closed fills and writes complete trade ledgers to state/trade_journal.json.
- Generates daily and weekly performance digests summarizing win rate, profit factor, and net PnL.

Agent 9: Adaptive Learning & Pattern Diagnosis

Source: multiagent_team/adaptive_learning.py

- Conducts an automated post-mortem audit after every closed trade.
- Assigns institutional letter grades (A+ to F) and self-scores execution discipline.
- Diagnoses exact root causes of wins and losses via feature attribution.
- Maintains persistent memory in pattern_learning_memory.json: awards +10 pt bonuses to winning patterns and blacklists failing ones.

Agent 10: Performance Coach & Operating Rules

Source: multiagent_team/performance_coach.py

- Tracks planned vs. actual execution variance and slippage drag.
- Monitors trailing win rates over 5-trade cohorts; triggers safety rollbacks if win rate drops below 20%.
- Maintains version-controlled active operating rules in state/active_operating_rules.json.

4. The Proprietary Strategy: Institutional Volatility-Liquidity (IVL) Absorption

The system's core edge is the **IVL (Institutional Volatility-Liquidity Absorption & Value Retest)** strategy. Rigorous backtesting across 1,920 15m historical candles on Delta Exchange confirmed **56.2% - 59.4% Win Rate, 1.53x - 1.71x Profit Factor, and +6.34R to +8.39R Net Profit** with 0.08R fee/slippage friction deducted per trade.

Rule Component	Mathematical Trigger & Logic	Purpose & Failure It Prevents
1. Coiled Volatility	ATR(14) ≤ 1.05 × SMA30(ATR)	Ensures energy is coiled before expansion; avoids buying into exhausted blowoff moves.
2. Volume Absorption	Volume ≥ 1.20 × SMA20(Volume)	Confirms whale and institutional accumulation rather than low-volume retail drift.
3. Liquidity Pivot Sweep	Price sweeps 10-12 bar swing extreme, closes back inside range	Exploits retail breakout traps; enters when market makers reverse after grabbing stop orders.
4. Absorption Wick	Lower wick ≥ 40% (Buy) or Upper wick ≥ 40% (Sell)	Proves aggressive price rejection and value retest.
5. Macro Confluence	EMA21/50 stack & Price vs. EMA200	Never trades against higher-timeframe momentum; aligns with 1H/4H trend.
6. Structural Noise Buffer	SL ≥ 0.35% for BTCUSD | SL ≥ 0.25% for ETHUSD (Max 0.65%)	Prevents normal 15m ATR random walk ($300-$500 on $80k BTC) from choking trades prematurely.

7. Milestone Trailing SL	At +0.8R: SL to -0.4R (cut risk 60%) At +1.1R: SL to +0.3R (lock green)	Eliminates breakeven stops while protecting capital and letting runners hit +2.5R to +2.6R target.

5. Execution Commands & Operational Guide

Operational Mode	Command Line	Description
Interactive Mode (Recommended)	`python main.py trade --symbols BTCUSD,ETHUSD --leverage 25`	Scans markets, calculates sizing, dispatches approval cards with inline buttons to Telegram and terminal before executing.
24/7 Scanner Loop	`python run_247_scanner.py --symbols BTCUSD,ETHUSD --interval 120`	Autonomous background loop executing periodic multi-timeframe scans, trailing stop updates, and state reconciliation.
One-Time Scan	`python main.py scan --symbols BTCUSD,ETHUSD`	Single inspection cycle printing market regime, liquidity sweeps, and qualified alpha setups without placing orders.
MCP Server	`python main.py mcp`	Launches the Model Context Protocol server exposing Delta Exchange tools to AI assistants (Claude, Cursor, Antigravity).
Daily Digest	`python main.py digest`	Generates an executive performance summary of completed trades, realized PnL, win rates, and fee drag.
Connectivity Test	`python main.py test`	Tests Delta Exchange API authentication, HMAC-SHA256 signature generation, and live wallet balances.

6. Safety Controls, Failsafes & State Files

The system persists all historical actions in human-readable JSON files in the state/ directory:

- **state/trade_journal.json**: Master ledger of every trade entry, exit price, R-multiple captured, and net realized PnL.
- **state/trailing_stop_state.json**: Real-time tracking of active positions, trailing stop stages, and parent/child order IDs.
- **state/pattern_learning_memory.json**: Agent 9's dynamic pattern memory, win/loss scores, and active pattern bonuses/penalties.
- **state/active_operating_rules.json**: Agent 10's version-controlled institutional operating rules and safety parameters.
- **state/system_state.json**: Portfolio health scores, daily drawdown counters, and circuit breaker status.

Built-In Failsafe Safeguards:

- **3.0% Max Daily Drawdown Circuit Breaker**: Halts all new orders immediately if daily losses reach 3.0% of portfolio equity.
- **Orphan Bracket Order Cleanup**: Automatically cancels leftover stop-loss or take-profit orders when a position closes, eliminating rogue fills.
- **Liquidation Armor**: Guarantees liquidation distance is ≥ 3x the stop-loss distance, preventing liquidation under all circumstances.

- **Emergency Killswitch**: Available in terminal or Telegram to instantly cancel all open orders and market-close all positions in 1 click.